

NAME

ovn-ctl – Open Virtual Network northbound daemon lifecycle utility

SYNOPSIS

ovn-ctl [*options*] *command* [-- *extra_args*]

DESCRIPTION

This program is intended to be invoked internally by Open Virtual Network startup scripts. System administrators should not normally invoke it directly.

COMMANDS

start_northd
start_controller
start_controller_vtep
start_ic
stop_northd
stop_controller
stop_controller_vtep
stop_ic
restart_northd
restart_controller
restart_controller_vtep
restart_ic
promote_ovnnb
promote_ovnsb
demote_ovnnb
demote_ovnsb
status_ovnnb
status_ovnsb
start_ovsdb
start_nb_ovsdb
start_sb_ovsdb
stop_ovsdb
stop_nb_ovsdb
stop_sb_ovsdb
restart_ovsdb
run_nb_ovsdb
run_sb_ovsdb
promote_ic_nb
promote_ic_sb
demote_ic_nb
demote_ic_sb
status_ic_nb
status_ic_sb
start_ic_ovsdb
start_ic_nb_ovsdb
start_ic_sb_ovsdb
stop_ic_ovsdb
stop_ic_nb_ovsdb
stop_ic_sb_ovsdb
restart_ic_ovsdb
run_ic_nb_ovsdb
run_ic_sb_ovsdb
start_ovnbr_ovsdb

```

start_br_controller
stop_ovnbr_ovsdb
stop_br_controller
restart_ovnbr_ovsdb
restart_br_controller
status_ovnbr_ovsdb
status_br_controller
run_ovnbr_ovsdb

```

OPTIONS

--ovn-northd-priority=*NICE* sets the ovn-northd process niceness. The default is **-10**.
--ovn-northd-wrapper=*WRAPPER* runs ovn-northd under the specified wrapper.
--ovn-controller-priority=*NICE* sets the ovn-controller process niceness. The default is **-10**.
--ovn-controller-wrapper=*WRAPPER* runs ovn-controller under the specified wrapper.
--ovn-ic-priority=*NICE* sets the ovn-ic process niceness. The default is **-10**.
--ovn-ic-wrapper=*WRAPPER* runs ovn-ic under the specified wrapper.
--ovnbr-controller-priority=*NICE* sets the ovn-br-controller process niceness. The default is **-10**.
--ovnbr-controller-wrapper=*WRAPPER* runs ovn-br-controller under the specified wrapper.
--ovsdb-nb-wrapper=*WRAPPER* runs the Northbound database server under the specified wrapper.
--ovsdb-sb-wrapper=*WRAPPER* runs the Southbound database server under the specified wrapper.
--ovsdb-ovnbr-wrapper=*WRAPPER* runs the OVN bridge controller database server under the specified wrapper.
--ovn-user=*USER[:GROUP]* runs OVN daemons as the specified user and optional group.
--ovn-manage-ovsdb=*yes|no* controls whether **start_northd** and **stop_northd** also manage the NB and SB database servers. The default is **yes**.
--ovn-controller-system-id=*SYSTEM_ID* sets the chassis name passed to ovn-controller. The default is empty.
--ovsdb-disable-file-column-diff=*yes|no* controls whether database servers are started with **--disable-file-column-diff**. The default is **no**.
-h | --help displays usage information.

FILE LOCATION OPTIONS

--db-sock=*SOCKET* sets the local Open vSwitch database socket.
--db-nb-file=*FILE* sets the Northbound database file.
--db-sb-file=*FILE* sets the Southbound database file.
--db-nb-sock=*SOCKET* and **--db-sb-sock=*SOCKET*** set the NB and SB database sockets. Their defaults are **OVN_RUNDIR/ovnnb_db.sock** and **OVN_RUNDIR/ovnsb_db.sock**, respectively.
--db-nb-schema=*FILE* sets the Northbound database schema file.
--db-sb-schema=*FILE* sets the Southbound database schema file.
--db-ic-nb-sock=*SOCKET* and **--db-ic-sb-sock=*SOCKET*** set the IC-NB and IC-SB database sockets. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.sock** and **OVN_RUNDIR/ovn_ic_sb_db.sock**, respectively.
 The **--db-nb-create-insecure-remote=*yes|no***, **--db-sb-create-insecure-remote=*yes|no***, **--db-ic-nb-create-insecure-remote=*yes|no***, **--db-ic-sb-create-insecure-remote=*yes|no***, and **--db-ovnbr-create-insecure-remote=*yes|no*** options accept **yes** or **no**. The default is **no**. When set to **yes**, the database server creates an insecure ptcp remote; **no** does not create that remote.

--db-ic-nb-file=FILE sets the IC-NB database file.
--db-ic-sb-file=FILE sets the IC-SB database file.
--db-ic-nb-schema=FILE sets the IC-NB database schema file.
--db-ic-sb-schema=FILE sets the IC-SB database schema file.
--db-nb-pidfile=FILE and **--db-sb-pidfile=FILE** set the NB and SB database pidfiles. Their defaults are **OVN_RUNDIR/ovnnb_db.pid** and **OVN_RUNDIR/ovnsb_db.pid**, respectively.
--db-nb-ctrl-sock=SOCKET and **--db-sb-ctrl-sock=SOCKET** set the NB and SB database control sockets. Their defaults are **OVN_RUNDIR/ovnnb_db.ctl** and **OVN_RUNDIR/ovnsb_db.ctl**, respectively.
--db-nb-config-file=FILE sets the Northbound database server configuration file.
--db-sb-config-file=FILE sets the Southbound database server configuration file.
--db-ic-nb-config-file=FILE sets the IC-NB database server configuration file.
--db-ic-sb-config-file=FILE sets the IC-SB database server configuration file.
--db-sb-relay-config-file=FILE sets the Southbound relay server configuration file.
--db-nb-use-remote-in-db, **--db-sb-use-remote-in-db**, **--db-ic-nb-use-remote-in-db**, and **--db-ic-sb-use-remote-in-db** accept **yes** or **no** and default to **yes**. A bare option is equivalent to specifying **yes**. The value **yes** makes ovnsdb-server listen on remotes configured in the database **connections** column, while **no** does not. If a corresponding configuration file is specified, it takes precedence and this setting is ignored.
--db-ovnbr-use-remote-in-db=yes|no controls whether ovnsdb-server listens on the remotes configured in the database **connections** column. The default is **yes**.
--ovn-controller-ssl-key=KEY sets the ovn-controller SSL/TLS private key file.
--ovn-controller-ssl-cert=CERT sets the ovn-controller SSL/TLS certificate file.
--ovn-controller-ssl-ca-cert=CERT sets the ovn-controller SSL/TLS CA certificate file.
--ovn-controller-ssl-bootstrap-ca-cert=CERT bootstraps the ovn-controller SSL/TLS CA certificate file.
--ovnbr-controller-ssl-key=KEY sets the ovn-br-controller SSL/TLS private key file.
--ovnbr-controller-ssl-cert=CERT sets the ovn-br-controller SSL/TLS certificate file.
--ovnbr-controller-ssl-ca-cert=CERT sets the ovn-br-controller SSL/TLS CA certificate file.
--ovnbr-controller-ssl-bootstrap-ca-cert=CERT bootstraps the ovn-br-controller SSL/TLS CA certificate file.
--ovn-ovnbr-db-ssl-key=KEY sets the ovn-br-controller database SSL/TLS private key file.
--ovn-ovnbr-db-ssl-cert=CERT sets the ovn-br-controller database SSL/TLS certificate file.
--ovn-ovnbr-db-ssl-ca-cert=CERT sets the ovn-br-controller database SSL/TLS CA certificate file.
--db-ovnbr-sock=SOCKET sets the OVN bridge controller database socket.
--db-ovnbr-file=FILE sets the OVN bridge controller database file.
--db-ic-nb-pidfile=FILE, **--db-ic-sb-pidfile=FILE**, and **--db-ovnbr-pidfile=FILE** set the IC-NB, IC-SB, and OVN bridge controller database pidfiles. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.pid**, **OVN_RUNDIR/ovn_ic_sb_db.pid**, and **OVN_RUNDIR/ovnbr_db.pid**, respectively.
--db-ic-nb-ctrl-sock=SOCKET, **--db-ic-sb-ctrl-sock=SOCKET**, and **--db-ovnbr-ctrl-sock=SOCKET** set database control sockets. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.ctl**, **OVN_RUNDIR/ovn_ic_sb_db.ctl**, and **OVN_RUNDIR/ovnbr_db.ctl**, respectively.

--db-ovnbr-schema=FILE sets the OVN bridge controller database schema file.

LOGGING OPTIONS

--ovn-controller-log=STRING, **--ovn-northd-log=STRING**, and **--ovn-ic-log=STRING** set daemon logging parameters. The default is **-vconsole:emer -vsyslog:err -vfile:info**.

--ovnbr-controller-log=STRING sets ovn-br-controller logging parameters. The default is **-vconsole:emer -vsyslog:err -vfile:info**.

--ovn-nb-log=STRING, **--ovn-sb-log=STRING**, **--ovn-ic-nb-log=STRING**, **--ovn-ic-sb-log=STRING**, **--ovn-sb-relay-log=STRING**, and **--ovn-ovnbr-log=STRING** set database-server logging parameters. The defaults for the NB, SB, IC-NB, IC-SB, and OVN bridge controller database servers are **-vconsole:off -vfile:info**. The default for the SB relay database server is **-vconsole:emer -vsyslog:err -vfile:info**.

--ovn-northd-logfile=FILE, **--ovn-ic-logfile=FILE**, **--ovn-nb-logfile=FILE**, **--ovn-sb-logfile=FILE**, **--ovn-ic-nb-logfile=FILE**, **--ovn-ic-sb-logfile=FILE**, **--ovn-sb-relay-logfile=FILE**, and **--ovn-ovnbr-logfile=FILE** select log files. Database-server log files default to files under **OVN_LOGDIR**; daemon log files default to empty.

--ovn-northd-n-threads=N_THREADS sets the number of ovn-northd threads. The default is **1**.

PROTOCOL, CIPHER AND CIPHERSUITE OPTIONS

The daemon and database prefixes also accept their SSL/TLS key, certificate, and CA certificate options. The prefixes are **ovn-controller**, **ovn-northd**, **ovn-ic**, **ovn-nb-db**, **ovn-sb-db**, **ovn-ic-nb-db**, **ovn-ic-sb-db**, **ovn-sb-relay-db**, and **ovn-ovnbr-db**; append **-ssl-key**, **-ssl-cert**, or **-ssl-ca-cert**. Controller prefixes also accept **-ssl-bootstrap-ca-cert**. Empty values leave the normal database or daemon TLS settings in effect.

--ovn-controller-ssl-protocols=PROTOCOLS.

--ovn-ic-ssl-protocols=PROTOCOLS.

--ovn-northd-ssl-protocols=PROTOCOLS.

--ovn-nb-db-ssl-protocols=PROTOCOLS.

--ovn-sb-db-ssl-protocols=PROTOCOLS.

--ovn-ic-nb-db-ssl-protocols=PROTOCOLS.

--ovn-ic-sb-db-ssl-protocols=PROTOCOLS.

--ovn-controller-ssl-ciphers=CIPHERS.

--ovn-ic-ssl-ciphers=CIPHERS.

--ovn-northd-ssl-ciphers=CIPHERS.

--ovn-nb-db-ssl-ciphers=CIPHERS.

--ovn-sb-db-ssl-ciphers=CIPHERS.

--ovn-ic-nb-db-ssl-ciphers=CIPHERS.

--ovn-ic-sb-db-ssl-ciphers=CIPHERS.

--ovn-controller-ssl-ciphersuites=CIPHERSUITES.

--ovn-ic-ssl-ciphersuites=CIPHERSUITES.

--ovn-northd-ssl-ciphersuites=CIPHERSUITES.

--ovn-nb-db-ssl-ciphersuites=CIPHERSUITES.

--ovn-sb-db-ssl-ciphersuites=CIPHERSUITES.

--ovn-ic-nb-db-ssl-ciphersuites=CIPHERSUITES.

--ovn-ic-sb-db-ssl-ciphersuites=CIPHERSUITES.

--ovn-controller-ssl-server-name=NAME.
--ovn-ic-ssl-server-name=NAME.
--ovn-northd-ssl-server-name=NAME.
--ovn-nb-db-ssl-server-name=NAME.
--ovn-sb-db-ssl-server-name=NAME.
--ovn-ic-nb-db-ssl-server-name=NAME.
--ovn-ic-sb-db-ssl-server-name=NAME.
--ovn-sb-relay-db-ssl-server-name=NAME.
--ovn-ovnbr-db-ssl-server-name=NAME.
--ovnbr-controller-ssl-protocols=PROTOCOLS sets the ovn-br-controller SSL/TLS protocols.
--ovnbr-controller-ssl-ciphers=CIPHERS sets the ovn-br-controller SSL/TLS cipher list.
--ovnbr-controller-ssl-ciphersuites=CIPHERSUITES sets the ovn-br-controller TLS 1.3 and later ciphersuite list.
--ovn-ovnbr-db-ssl-protocols=PROTOCOLS sets the ovn-br-controller database SSL/TLS protocols.
--ovn-ovnbr-db-ssl-ciphers=CIPHERS sets the ovn-br-controller database SSL/TLS cipher list.
--ovn-ovnbr-db-ssl-ciphersuites=CIPHERSUITES sets the ovn-br-controller database TLS 1.3 and later ciphersuite list.

ADDRESS AND PORT OPTIONS

--db-nb-sync-from-addr=IP ADDRESS.
--db-nb-sync-from-port=PORT NUMBER.
--db-nb-sync-from-proto=PROTO.
--db-sb-sync-from-addr=IP ADDRESS.
--db-sb-sync-from-port=PORT NUMBER.
--db-sb-sync-from-proto=PROTO.
--db-ic-nb-sync-from-addr=IP ADDRESS.
--db-ic-nb-sync-from-port=PORT NUMBER.
--db-ic-nb-sync-from-proto=PROTO.
--db-ic-sb-sync-from-addr=IP ADDRESS.
--db-ic-sb-sync-from-port=PORT NUMBER.
--db-ic-sb-sync-from-proto=PROTO.

The options **--db-nb-addr**, **--db-nb-port**, **--db-sb-addr**, **--db-sb-port**, **--db-ic-nb-addr**, **--db-ic-nb-port**, **--db-ic-sb-addr**, **--db-ic-sb-port**, **--db-ovnbr-addr**, and **--db-ovnbr-port** select the addresses and ports used when the corresponding **--db-*-create-insecure-remote** option is **yes**. When that option is **no**, these settings have no effect. The defaults for **--db-nb-addr** and **--db-nb-port** are **0.0.0.0** and **6641**, respectively. The defaults for **--db-sb-addr** and **--db-sb-port** are **0.0.0.0** and **6642**, respectively. The defaults for **--db-ic-nb-addr** and **--db-ic-nb-port** are **0.0.0.0** and **6645**, respectively. The defaults for **--db-ic-sb-addr** and **--db-ic-sb-port** are **0.0.0.0** and **6646**, respectively. The defaults for **--db-ovnbr-addr** and **--db-ovnbr-port** are **0.0.0.0** and **6651**, respectively.

--ovn-northd-nb-db and **--ovn-northd-sb-db** select the NB and SB remotes used by ovn-northd. Similarly, **--ovn-ic-nb-db** and **--ovn-ic-sb-db** select the IC database remotes. Their defaults are the corresponding local unix database sockets.

--ovn-northd-nb-db=PROTO:IP ADDRESS: PORT..

--ovn-northd-sb-db=PROTO:IP ADDRESS: PORT..

--ovn-ic-nb-db=PROTO:IP ADDRESS: PORT..

--ovn-ic-sb-db=PROTO:IP ADDRESS: PORT..

CLUSTERING OPTIONS

--db-nb-cluster-local-addr=IP ADDRESS.

--db-nb-cluster-local-port=PORT NUMBER.

--db-nb-cluster-local-proto=PROTO (tcp/ssl).

--db-nb-cluster-remote-addr=IP ADDRESS.

--db-nb-cluster-remote-port=PORT NUMBER.

--db-nb-cluster-remote-proto=PROTO (tcp/ssl).

--db-nb-election-timer=Timeout in milliseconds.

--db-sb-cluster-local-addr=IP ADDRESS.

--db-sb-cluster-local-port=PORT NUMBER.

--db-sb-cluster-local-proto=PROTO (tcp/ssl).

--db-sb-cluster-remote-addr=IP ADDRESS.

--db-sb-cluster-remote-port=PORT NUMBER.

--db-sb-cluster-remote-proto=PROTO (tcp/ssl).

--db-sb-election-timer=Timeout in milliseconds.

--db-ic-nb-cluster-local-addr=IP ADDRESS.

--db-ic-nb-cluster-local-port=PORT NUMBER.

--db-ic-nb-cluster-local-proto=PROTO (tcp/ssl).

--db-ic-nb-cluster-remote-addr=IP ADDRESS.

--db-ic-nb-cluster-remote-port=PORT NUMBER.

--db-ic-nb-cluster-remote-proto=PROTO (tcp/ssl).

--db-ic-sb-cluster-local-addr=IP ADDRESS.

--db-ic-sb-cluster-local-port=PORT NUMBER.

--db-ic-sb-cluster-local-proto=PROTO (tcp/ssl).

--db-ic-sb-cluster-remote-addr=IP ADDRESS.

--db-ic-sb-cluster-remote-port=PORT NUMBER.

--db-ic-sb-cluster-remote-proto=PROTO (tcp/ssl).

--db-cluster-schema-upgrade=yes|no controls whether clustered database servers automatically upgrade their schemas when they start. The default is **yes**; specify **no** to disable the automatic upgrade.

--no-db-cluster-schema-upgrade is equivalent to **--db-cluster-schema-upgrade=no**. The default is **yes**.

PROBE INTERVAL OPTIONS

--db-nb-probe-interval-to-active=Time in milliseconds .

--db-sb-probe-interval-to-active=Time in milliseconds .

RELAY OPTIONS

--db-sb-relay-sock=SOCKET, **--db-sb-relay-pidfile=FILE**, and **--db-sb-relay-ctrl-sock=SOCKET** set the relay socket, pidfile, and control socket. The defaults are **ovnsb_relay_db.sock**, **ovnsb_relay_db.pid**, and **ovnsb_relay_db.ctl** in **OVN_RUNDIR**.

--db-sb-relay-remote=REMOTE selects the upstream relay remote. The default is empty.

--db-sb-relay-use-remote-in-db=yes|no controls whether ovsdb-server listens on the Southbound remotes configured in the database **connections** column. The default is **yes**. A bare option is equivalent to specifying **yes**.

EXTRA OPTIONS

Any options after '-' will be passed on to the binary run by *command* with the exception of *start_northd*, which can have options specified in *ovn-northd-db-params.conf*. Any *extra_args* passed to *start_northd* will be passed to the ovsdb-servers if **--ovn-manage-ovsdb=yes**

CONFIGURATION FILES

Following are the optional configuration files. If present, it should be located in the etc dir

ovnnb-active.conf

If present, this file should hold the url to connect to the active Northbound DB server

tcp:x.x.x.x:6641

ovnsb-active.conf

If present, this file should hold the url to connect to the active Southbound DB server

tcp:x.x.x.x:6642

ovn-northd-db-params.conf

If present, *start_northd* will not start the DB server even if **--ovn-manage-ovsdb=yes**. This file should hold the database url parameters to be passed to *ovn-northd*.

--ovnnb-db=tcp:x.x.x.x:6641 --ovnsb-db=tcp:x.x.x.x:6642

ic-nb-active.conf

If present, this file should hold the url to connect to the active Interconnection Northbound DB server

tcp:x.x.x.x:6645

ic-sb-active.conf

If present, this file should hold the url to connect to the active Interconnection Southbound DB server

tcp:x.x.x.x:6646

ovn-ic-db-params.conf

If present, this file should hold the database url parameters to be passed to *ovn-ic*.

--ic-nb-db=tcp:x.x.x.x:6645 --ic-sb-db=tcp:x.x.x.x:6646

RUNNING OVN DB SERVERS WITHOUT DETACHING

ovn-ctl run_nb_ovsdb

This command runs the OVN nb ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN nb ovsdb-server in a container.

ovn-ctl run_sb_ovsdb

This command runs the OVN sb ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN sb ovsdb-server in a container.

ovn-ctl run_ic_nb_ovsdb

This command runs the OVN IC-NB ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN IC-NB ovsdb-server in a container.

ovn-ctl run_ic_sb_ovsdb

This command runs the OVN IC-SB ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN IC-SB ovsdb-server in a container.

ovn-ctl run_ovnbr_ovsdb

This command runs the OVN bridge db `ovsdb-server` without passing the **detach** option, making it to block until `ovsdb-server` exits. This command will be useful for starting the OVN br db `ovsdb-server` in a container.

EXAMPLE USAGE

Run `ovn-controller` on a host already running OVS

```
# ovn-ctl start_controller
```

Run `ovn-northd` on a host already running OVS

```
# ovn-ctl start_northd
```

All-in-one OVS+OVN for testing

```
# ovs-ctl start --system-id="random"
```

```
# ovn-ctl start_northd
```

```
# ovn-ctl start_controller
```

Promote and demote `ovsdb` servers

```
# ovn-ctl promote_ovnnb
```

```
# ovn-ctl promote_ovnsb
```

```
# ovn-ctl --db-nb-sync-from-addr=x.x.x.x --db-nb-sync-from-port=6641 --db-nb-probe-interval-to-active=60000 demote_ovnnb
```

```
# ovn-ctl --db-sb-sync-from-addr=x.x.x.x --db-sb-sync-from-port=6642 --db-sb-probe-interval-to-active=60000 demote_ovnsb
```

Creating a clustered db on 3 nodes with IPs `x.x.x.x`, `y.y.y.y` and `z.z.z.z`

Starting OVN `ovsdb` servers and `ovn-northd` on the node with IP `x.x.x.x`

```
# ovn-ctl --db-nb-addr=x.x.x.x --db-nb-create-insecure-remote=yes --db-sb-addr=x.x.x.x
--db-sb-create-insecure-remote=yes --db-nb-cluster-local-addr=x.x.x.x --db-sb-cluster-local-addr=x.x.x.x
--ovn-northd-nb-db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641
--ovn-northd-sb-db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd
```

Starting OVN `ovsdb`-servers and `ovn-northd` on the node with IP `y.y.y.y` and joining the cluster started at `x.x.x.x`

```
# ovn-ctl --db-nb-addr=y.y.y.y --db-nb-create-insecure-remote=yes --db-sb-addr=y.y.y.y
--db-sb-create-insecure-remote=yes --db-nb-cluster-local-addr=y.y.y.y --db-sb-cluster-local-addr=y.y.y.y
--db-nb-cluster-remote-addr=x.x.x.x --db-sb-cluster-remote-addr=x.x.x.x
--ovn-northd-nb-db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641
--ovn-northd-sb-db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd
```

Starting OVN `ovsdb`-servers and `ovn-northd` on the node with IP `z.z.z.z` and joining the cluster started at `x.x.x.x`

```
# ovn-ctl --db-nb-addr=z.z.z.z --db-nb-create-insecure-remote=yes --db-nb-cluster-local-addr=z.z.z.z
--db-sb-addr=z.z.z.z --db-sb-create-insecure-remote=yes --db-sb-cluster-local-addr=z.z.z.z
--db-nb-cluster-remote-addr=x.x.x.x --db-sb-cluster-remote-addr=x.x.x.x
--ovn-northd-nb-db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641
--ovn-northd-sb-db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd
```

Passing ssl keys when starting OVN dbs will supersede the default ssl values in db

Starting standalone ovn db server passing SSL/TLS certificates

```
# ovn-ctl --ovn-nb-db-ssl-key=/etc/ovn/ovnnb-privkey.pem
--ovn-nb-db-ssl-cert=/etc/ovn/ovnnb-cert.pem --ovn-nb-db-ssl-ca-cert=/etc/ovn/cacert.pem
--ovn-sb-db-ssl-key=/etc/ovn/ovnsb-privkey.pem --ovn-sb-db-ssl-cert=/etc/ovn/ovnsb-cert.pem
--ovn-sb-db-ssl-ca-cert=/etc/ovn/cacert.pem start_northd
```

Avoiding automatic clustered OVN database schema upgrade

If you desire more control over clustered DB schema upgrade, you can opt-out of automatic on-start upgrade attempts with **--no-db-cluster-schema-upgrade**.

Start OVN NB and SB clustered databases on host with IP x.x.x.x without schema upgrade

```
# ovn-ctl start_nb_ovsdb --db-nb-cluster-local-addr=x.x.x.x --no-db-cluster-schema-upgrade
# ovn-ctl start_sb_ovsdb --db-sb-cluster-local-addr=x.x.x.x --no-db-cluster-schema-upgrade
```

Trigger clustered DB schema upgrade manually

```
# ovsdb-client convert unix:/var/run/ovn/ovnnb_db.sock /usr/local/share/ovn/ovn-nb.ovsschema #
ovsdb-client convert unix:/var/run/ovn/ovnsb_db.sock /usr/local/share/ovn/ovn-sb.ovsschema
```

Run OVN bridge controller services on a host already running OVS

```
# ovn-ctl start_ovnbr_ovsdb
# ovn-ctl start_br_controller
```